
Power-Centric Model of Risk & Disambiguating Societal Power-Dynamics with Concrete Economic and Mathematical Models

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Abstract

Power-Centric Framework of Risk. I propose a power-centric framework for understanding AI risks, arguing that catastrophic outcomes – from gradual disempowerment to authoritarian control – stem from common power-transfer mechanisms rather than purely technical failures. Using evolutionary game theory, I formalize how AI creates self-reinforcing dynamics that: (1) incentivize initial participation, (2) generate collective harm, (3) erect high exit barriers, and (4) ossify power structures.

Formalization of Power. The motivation here is being able to describe and aggregate dangerous power dynamics between populations with complicated relationships including ones that can't quite be assigned a utility function over long periods of time where people perhaps aren't acting rationally. Starting from basic models of power and continue breaking things to demonstrate the required mathematical properties a model of power must have.

1. Logic of Catastrophe

Not remotely started, need to build from scratch. Might be better to bring into scope after power.

1.1. Modeling States/Actors/Environment

$$s \in \text{States} \quad (1)$$

$$x \in X \text{ (actors = people, inanimate object, environment)} \quad (2)$$

environment, $e \in X$

(One actor, essentially representing 'everything else')⁽³⁾

$$\forall s \in S, \exists x, y.$$

$x \neq y \wedge x, y \in s$ (ie: All states have two unique actors)⁽⁴⁾

$\forall s \in S. e \in s$ (ie: Environment is in all states) ⁽⁵⁾

Logical Relationships & Progression of Claims Power Volume and Danger

- Little power $\Rightarrow$ little effects $\Rightarrow$ localized & manageable danger
- Large danger $\Rightarrow$ large power

Dynamics

- Balance of power $\wedge$ agents $\Rightarrow$ equilibrium
- Equilibrium $\neq$ no danger
- Equilibrium $\wedge$ something(?) $\Rightarrow$ no danger

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2. Sociological Definition of Power

- Dahlian/Pluralist Power (M) – A can get B to do something that B would not otherwise do (more rooted in the ability to get some boolean event to happen when they do/don't take some action.)
 - Capability (Could plausibly do it)
 - Control (Would do it, utility based)
- French and Raven
 - Reward Power – A 's perception that B has the ability to mediate rewards for him
 - Coercive Power – A 's perception that B has ability to mediate punishments for him (do something against their will)
- Luke's dimensions of power (more studying how decisions get made than any formal definitions of power.)

Combining the above, we will say:

- **Power** of A over B is the ability for A to determine the rewards/punishments of B (Independent of recognition which is dissimilar to French/Raven)

 Talk about in more detail. Just general background on power. This should perhaps come first in the paper.

3. Towards a Formal Model of Power

Motivation. There are many definitions of power across sociological, economic, political, and anthropological literature. However, almost none of these definitions were formalized, each disagreeing with and pointing out flaws in the definitions of power offered by their predecessors, offering new ones in their place. However, many of these definitions of power can be related to one another. If we are able to distinguish between these different types of power and cut through the different assumptions their usages have, we can have a much clearer conversation on the topic, potentially clear the way towards empirically measuring the potential and manifest power at large in our society, and design mechanisms to avoid harmful power dynamics.¹

Practicality. Far from being just a philosophical exercise, there is interest in modelling power, values, and alignment between agents in other domains. Two examples: (a) In Reinforcement Learning, a core subdomain of AI research, there is interest in understanding whether two agents are pursuing the same goals even when pursuing different actions. (b) Researchers at the Centre for the Governance for AI are interested in formalizing how the social contract can break down when labor replacing AI is introduced in addition to interest in formalizing the conversion of economic power into political power and vice versa. While this work may not bring us quite that far, it at least lays the groundwork for what can hopefully be a larger framework.

Scope. Rather than create a universal “Theory of Power” encompassing all definitions, I begin by developing a concrete mathematical framework for one specific perspective on power. We will either:

- (a) Formally relate other definitions to this framework
- (b) Demonstrate their incoherence
- (c) Identify behaviorally irrelevant aspects (philosophical rather than practical²)
- (d) Show where our model reasonably approximates complex realities

Goal. Although this aims to develop a general theory of power, the shape this will take will inevitably be influenced by context in which it was developed. This formalism was developed in the context of understanding and predicting dangerous power dynamics by describing and comparing

power dynamics between populations with complex relationships – without assuming rationality and sometimes without clear utility functions – over a long period of time.

“Thus we are not likely to produce certainly not for some considerable time to come anything like a single, consistent, coherent ‘Theory of Power.’ We are much more likely to produce a variety of theories of limited scope, each of which employs some definition of power that is useful in the context of the particular piece of research or theory but different in important respects from the definitions of other studies.”

— Dahl's Concept of Power 1957

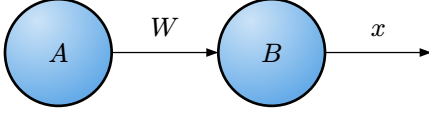
Methodology. I start with one of the original general formalisms of power in literature – Dahl's influential essay on the Concept of Power from 1957. I attempt to generalize this even further, point out some of the issues of it, demonstrate the mathematical properties a model of power must have, and continue adding components until we are closer to modeling what we care about.

¹Existing theories of power are either loosely connected and non-rigorously defined OR rigorously mathematically defined in the context of a particular limited model – for example, Cooperative Game Theory has multiple indices of power, but these cannot be easily extended outside the realm of these models. Leaving out details you don't care about is a feature not a bug. However, it can make your models brittle when moved outside the topic.

²Akin to philosophy of quantum mechanics

4. Generalized Dahl Framework

4.1. Defining Dahlian Power



I start with Dahl’s influential concept, which I term **Dahlian Power** to distinguish it from other power definitions we’ll encounter.

Dahl’s Intuition. Consider two scenarios from the original essay³

1. A person on a street corner “commanding” drivers to use the right side (which they already do)
2. A police officer directing traffic to turn when it would normally go straight.

Only the second demonstrates genuine power – the ability to cause behavior that wouldn’t otherwise occur.

Formal Definition: Dahlian Power of agent A ’s control over B with respect to the response x by means w :

$$M\left(\frac{A}{B} : W, x\right) := \Pr(B, x \mid A, w) - \Pr(B, x \mid A, \bar{w}) \quad (6)$$

Where:

- $\Pr(B, x \mid A, w)$ is the probability that B takes action x given A took action w .
- $\bar{w}$ is A not taking action w

Key Components. Dahl’s model is made up of the following components:

1. **Agents.** At least two agents each of which have binary **actions** (e.g. take action w or not)

³The full quote:

“Suppose I stand on a street corner and say to myself, ‘I command all automobile drivers on this street to drive on the right side of the road’; suppose further that all the drivers actually do as I ‘command’ them to do; still, most people will regard me as mentally ill if I insist that I have enough power over automobile drivers to compel them to use the right side of the road. On the other hand, suppose a policeman is standing in the middle of an intersection at which most traffic ordinarily moves ahead; he orders all traffic to turn right or left; the traffic moves as he orders it to do. Then it accords with what I conceive to be the bedrock idea of power to say that the policeman acting in this particular role evidently has the power to make automobile drivers turn right or left rather than go ahead. My intuitive idea of power, then, is something like this: A has power over B to the extent that he can get B to do something that B would not otherwise do. – Dahl 1957”

2. **Connection.** Some mechanism J linking agents (unspecified)
3. **Probability.** Distribution over agent-action pairs

What this doesn’t require:

1. **Utility.** Doesn’t say what either agent prefers.
2. **Causal Relationship.** Dahl explicitly avoids requiring causal traces – only observable behavioral differences matter. However, does note causality is required for real power, hence the means must precede the response temporally.
3. **Practicality.** Doesn’t say whether the controller actually would or would not take action w in practice.

4.2. Generalizing Beyond Binary Actions

Dahl’s binary framework (action taken or not) is limiting. Real agents choose from multiple options.

Extension to n actions. Let $\mathcal{W} := \{w_1, \dots, w_n \mid n \in \mathbb{N}^+\}$. Then:

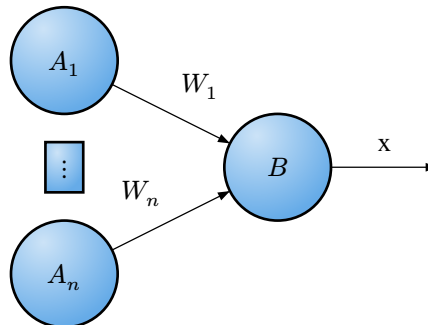
$$M\left(\frac{A}{B} : W, x\right) := \max_{w^+ \in \mathcal{W}} \Pr(B, x \mid A, w^+) - \max_{w^- \in \mathcal{W}} \Pr(B, x \mid A, w^-) \quad (7)$$

This measures A ’s maximum influence over B ’s likelihood of taking action x .

Boundary Case. When $|\mathcal{W}| = 1$ (no choice), Dahlian Power $M = 0$, fitting our intuition that power requires agency.

4.3. Beyond Two-Controller Dynamics

Real power dynamics rarely involve just two controllers. Consider the example from Dahl’s original paper where multiple legislators influence the outcome of a policy being passed or not:



When multiple controllers $\{A_1, \dots, A_n\}$ simultaneously influence the responder B , we need to extend our framework. Each agent A_i can take actions from their action set $\mathcal{W}_i$, creating a joint action profile $w = (w_1, \dots, w_n)$.

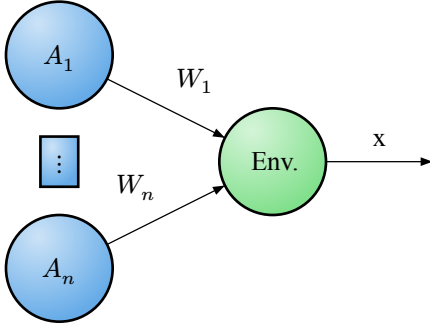
Comparative Dahlian Power. To assess individual contributions within multi-controller influence, Dahl introduced comparative measures (Referred to as M'' and did not formalize). For controller A_i within controllers A :

■ Finish, not super important.

$$M_i\left(\frac{A}{B} : \mathcal{W}, x\right) := \max_{w \in \prod \mathcal{W}_j} \dots \quad (8)$$

Combined Power. We can aggregate any combination of these controllers into a virtual single controller coalition $A' \subseteq \{A_1, \dots, A_n\}$ and define the power of the coalition in a similar way. This new coalition A' will have an action space constructed from actions spaces of it's members $\mathcal{W}' = \mathcal{W}_1 \times \mathcal{W}_2 \times \dots \times \mathcal{W}_m$ where $A_1, \dots, A_m$ are the controllers in the coalition. ■ Better written def, (not necessarily referring to the indices of the original ordering)

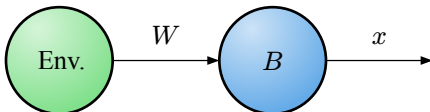
4.4. The Environment as a Black Box



Abstracting Away Agency. Dahl's framework doesn't require the influenced party to be an agent. By treating B 's decision-making as a black box – encoded entirely in the probability function $\Pr$ – we can extend Dahlian Power to environmental outcomes. Not only is this possible, but it may actually be more proper because Dahl doesn't model B as an agent taking actions at all. Letting $E :=$ environment

$$M\left(\frac{A}{E} : \mathcal{W}, x\right) := \max_{w^+ \in \mathcal{W}} \Pr(E = x \mid A, w^+) - \max_{w^- \in \mathcal{W}} \Pr(E = x \mid A, w^-) \quad (9)$$

Reverse Environmental Influence.



Can environments exert “power”? While environments don't take deliberate actions, environmental states do influence agent behavior:

Example. Weather “determines” whether others will attend your picnic – rain reduces attendance probability. While this stretches our definition of power depending on how you view the environment, it may still be sensible to develop analogues:

- Possible Environmental States = Action Space

The Agency Question. Dahlian power doesn't require desires or utility functions – only the ability to select from multiple actions. This creates a philosophical tension:

- Deterministic view: If environments (or agents) have no genuine choice, then $|\mathcal{W}| = 1$, yielding zero Dahlian power
- Probabilistic view: We assign probabilities to environmental states due to our uncertainty, treating nature as if it “chooses” stochastically

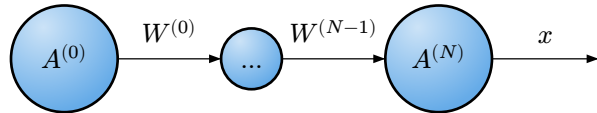
Practical Resolution. Whether discussing weather patterns or legislative decisions, we model both agents and environments probabilistically because:

- Complete determinism is computationally intractable
- Uncertainty is inherent to our observations
- Probabilistic models yield useful predictions

This is a methodological choice, not an ontological claim about free will or determinism. We use whichever representation – agent or environment – best serves our analytical goals.

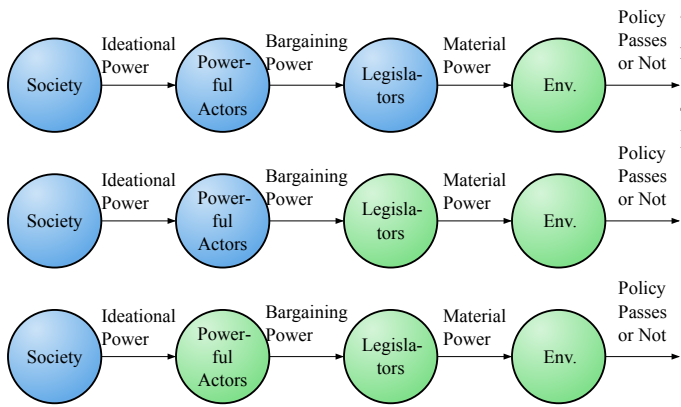
4.5. Means-Response Chains

Means-response chains DON'T WORK IN DAHL'S MODEL!!



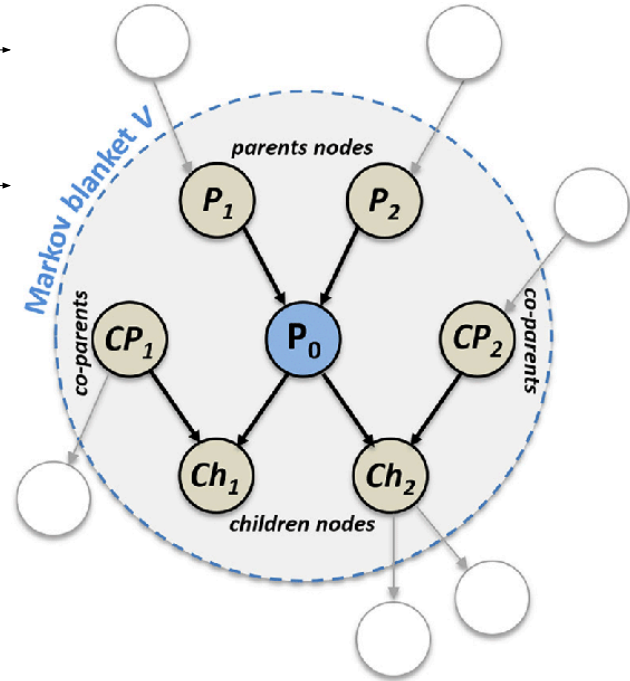
■ Actually this seems like it might not be possible as the model does not account for the inability to choose an available action!!! Can bring in the need for a behavioral approach to control happening later on. Instantly connects with some version of game theory(?). Only works if this is the environment. Actually no, works for players as well. But takes this really weird naïve stance on power that assigns more agency to a person than they actually have. This is where we can say there is a separation between power and control. (Ex: Action to burn down the entire Senate.) Best called Material power?

Also this doesn't seem to be doing some kind of attribution of control properly. Capability never disappears.



Policy $w = (w_1, \dots, w_n)$. Each of which interact through some mechanism modeled by Pr to affect the response x .

4.7. Generalizing Dahl to Markov Nets, HMMs, and MDPs



Lukes criticizes Dahl and the pluralists for being too empirical, only examining the political power of observable direct decision making, whereas there are plenty of soft-power leading up to that direct decision making that play a part in both (a) Shaping the preferences of voting and policies brought to the table prior to receiving the policy (bargaining power) (b) A more cultural notion of which ideas are even consciously considered (ideational power)

check + understand better Perhaps fun to dig up Dahl's New Haven data and demonstrate some type of this modeling. This categorization is extremely helpful for analyzing power and it will be returned to. However, can demonstrate mathematically what Luke's criticism is – ie: Doesn't really tell you who has the final say.

Even when modeling the potential actors that could be influencing the final legislator decisions, you come to the point of having to model all downstream agents as just “part of the environment”, not modeling any of the complexities and interactions of the underlying distribution. Additionally, modeling any of the upstream agents and asking how much power they have over the final decision requires modeling the rest of the agents as part of the environment. Regarding who your controller-in-power is.

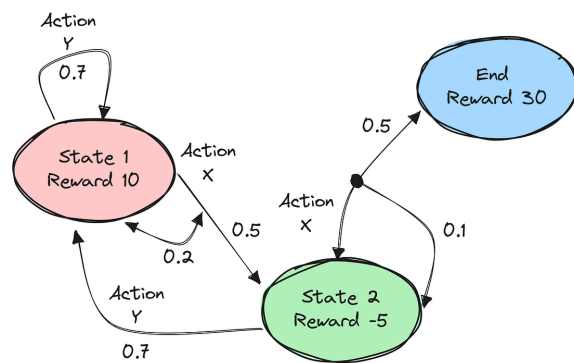
Additionally, you could model each of these actors as being individual controllers

Immediately this has some connection with law, duress, consent, etc. It seems like the actual utility is an important question here, ex: no one expects positive rewards to hold up in court, but coercion does. This in some way connects with consent.

Ideological power is somehow both an information asymmetry and an alteration to one's payoff functions.

4.6. Complexity of Dahl and Need For Compact Representations

In this formalism of power, for N controlling player and a max of M actions for each, there are $N \cdot M$ action profiles



In broad strokes, Dahl's theory of power is similar to the rich literature of markov nets in computer science literature. With a few differences:

- Taken from the perspective of actions that players take. (However these can be mapped onto priors in a markov net so long as there are no loops, which of course there are. Might not actually be an issue.)
- Control must be causal. Since this is near-impossible to prove in-full, it suffices to say that it's certainly not causal if the response happens before the action.

A few nice properties and intuitions can however be carried over:

- Markov blankets (You can isolate control by)
- Can be mapped to time-series HMMs.

Notoriously very difficult.  Maybe also show some existing CS problem to apply control in a new context.  Also maybe see what other studies of power exist out there that looked at causal relationships in this way. Certainly not a new understanding of power, just one that wasn't looked at broadly.

Already we are at a point where we can define control in MDPs, one of the most fundamental structures in reinforcement learning and applicable to many scenarios.  Can we apply this to some textbook MDP scenario?  Not sure I should transition to MDPs without bringing in utility as a pre-requisite of power.

4.8. Connecting Dahl to Game Theory

What Dahl's theory of power does not say is HOW the players in this scenario determine their actions, keeping the probability function $\Pr$ doing a lot of work here.

4.9. Criticisms of Dahl

This is not to overly simplify these models, only to ground them. Much more complex relationship with loops going back on itself over time.

 Complete.

5. Why Actions are Necessary for Power Analysis

5.1. The Inseparability Thesis

Claim. Power is meaningless without considering available actions – these elements are fundamentally inseparable in any coherent power framework.

5.1.1. Behavioral Power

Insight 1. Power is a function of available actions.

Insight 2. Power is a function of optimal actions assuming rationality.

I might name this control.

▢ Finish.

6. Why Absolute Power Fails to Model Power Dynamics

▢ Should this only be possible once utility is introduced?

▢ From cooperative game theory, we have weighted graphs, rule-based representations, and weighted-voting games.

6.1. The Modeling Asymmetry Thesis

Claim. Models of relative power ($\mathcal{R}$) form a proper superset of models of absolute power ($\mathcal{A}$), establishing that $\mathcal{R}$ is more expressive than $\mathcal{A}$ in modeling capacity.

Dahl says that there must be a “connection” for there to be a power dynamic.

6.1.1. Formal Statement

Definition (Power Model Spaces). Let:

- $\mathcal{A} = \{M :$
 $M \text{ models power as properties of agents}\}$
- $\mathcal{R} = \{M :$
 $M \text{ models power as relational}$
 $\text{properties between agents}\}$

Then: $\mathcal{A} \subset \mathcal{R}$ (proper subset relation)

6.1.2. Supporting Arguments

Theorem (Reduction Principle). Any absolute power model can be reformulated as a relative power model by introducing an environmental baseline.

Formal Construction. Given absolute power $P_{\text{abs}} : \text{Agent} \rightarrow \mathbb{R}^+$, we can define:

$$P_{\text{rel}(a,e)} = P_{\text{abs}(a)} - P_{\text{abs}(e)} \quad (10)$$

where $P_{\text{abs}(e)}$ serves as a zero-point reference for environment e .

▢ Replace with an actual proof from graph theory

Physical Analogy. This construction mirrors potential energy in physics, where absolute potential V is meaningless without a reference frame:

$$V_{\text{rel}} = V_{\text{abs}} - V_{\text{ground}} \quad (11)$$

Incompleteness of Absolute Power Models ($\mathcal{A} \subsetneq \mathcal{R}$).

Theorem (Limitation Theorem). Absolute power models cannot capture essential relational dynamics without introducing factors outside their framework.

▢ Demonstrate some examples like in my original, such as distance as one factor and just other seemingly complex set of environmental interaction factors that seem almost impossible to resolve absolute powers between entities to relative powers in a sensible way. In physics potential can be mapped to potential energy using distance, unlikely we have anything nearly as similar.

Core Problem. The power relationship between agents A and B cannot be logically derived from their absolute powers alone. Power lacks transitivity – if A dominates B and B dominates C , this does not guarantee A dominates C . This non-transitivity demonstrates that power cannot be reduced to a single absolute measure.

Example. Consider a circular dominance structure (like rock-paper-scissors):

- Rock defeats Scissors
- Scissors defeats Paper
- Paper defeats Rock

No absolute power assignment P_{abs} can capture these relationships.

Conclusion. Since capturing these dynamics requires factors $\notin \mathcal{A}$, we have $\mathcal{A} \neq \mathcal{R}$.

▢ improve

6.1.3. Proof of Proper Subset Relation

From §1.2.1: $\mathcal{A} \subseteq \mathcal{R}$

From §1.2.2: $\exists M \in \mathcal{R}$ such that $M \notin \mathcal{A}$

Therefore: $\mathcal{A} \subset \mathcal{R}$ (proper subset) $\square$

6.1.4. Power Over Environment as Logical Encapsulation of What is Meant By “Absolute Power”

📄 fin

6.1.5. Conclusion

Absolute power alone is insufficient for modeling power. For the remainder of this work, we consider only relational power – power over the environment or power over other agents.

Note: This does NOT tell us that relative power is sufficient, only that it may be.

7. Why Utility is Necessary for Power Analysis

7.1. The Inseparability Thesis

Claim. Power is meaningless without considering agent utilities – these elements are fundamentally inseparable in any coherent power framework.

NOTE: Will likely replace this utility statement.

7.1.1. Utility Alignment

Core Insight. Power is an incoherent concept under complete alignment of utilities.

Definition (Utility Alignment). Let:

- $U_A, U_B : \mathcal{S} \rightarrow \mathbb{R}$ be utility functions for agents A and B
- $\mathcal{S}$ be the state space
- $P(A \rightarrow B)$ be the power of A over B

If $U_A(s) \propto U_B(s) \forall s \in \mathcal{S}$ (perfect alignment), then $P(A \rightarrow B)$ is behaviorally irrelevant.

Illustrative Example. Consider a human-loving agent who can travel to an island, harm anywhere from 1 to 1,000,000 innocent people, and then escape unnoticed. Does their power increase with the number they *could* harm? This capacity is irrelevant:

- The human-loving agent doesn't benefit from causing harm and would prefer not to
- The potential victims have no desire to be harmed

The mere existence of this action is meaningless without considering whether it would ever be exercised.

 Replace with better example.

 Include behavioral misalignment, ie: power over

7.1.2. Agent Boundary Definition

Theorem (Aggregation Principle). Alignment relationships determine when individual power can be meaningfully aggregated into a single coherent agent:

$$\left[P_{\text{collective}} = \sum_i P_i \right] \Leftrightarrow [U_i \approx U_j \forall i, j] \quad (12)$$

Application. Humanity's collective power is discussable precisely because humans share sufficient utility alignment over key domains.

Connection to Sociology This is akin to *power-within* discussed in sociological literature.

7.1.3. Environmental Mediation

Extended Framework. When modeling multi-agent systems with environmental interactions:

- Environment E can be treated as an agent with its own dynamics
- Even without direct conflict between agents, they may compete for environmental resources
- Agents may inflict harm indirectly through environmental manipulation

Power becomes relevant when agents compete for control over shared environmental factors, even if their ultimate goals don't directly conflict.

 Also could say that their actions are some mixed strategy over a number of actions. Not sure if this is true but not a crucial point.

7.1.4. Agents in Shared State Space

Agents of course, are themselves components of state spaces and they may have preferences over the state of themselves and others. Often times they have preferences for their survival which must be modeled in the utility function as well.

7.1.5. Critique of Light-Cone Formulation

Orthogonality Scenario. Consider alien civilization Ω with utilities orthogonal to humanity H . Imagine Ω cares only about arranging neutrinos into smiley faces – an activity that has zero impact on human flourishing. Meanwhile, humans optimize matter and energy arrangements with no effect on neutrino patterns. (We can additionally imagine that neither agent meaningfully interferes with the other, perhaps humanity and these aliens never interact)

Even if Ω achieves perfect control over neutrino arrangements while humans have only partial control over matter/energy, it's meaningless to compare their "power over the environment." The concept of environmental control is incoherent without specifying *which aspects* of the environment matter.

Key Insight. Power must be defined over relevant state space – control over factors that actually matter to the agents involved:

$$P_{\text{relevant}(A)} = \int_{\mathcal{S}_A} |\nabla U_A(s)| \, ds \quad (13)$$

where $\mathcal{S}_A := \{s \in \mathcal{S} : \partial U_A / \partial s \neq 0\}$ represents the relevant state space.

Note that this relevant state space is inherently chosen in reference to some agent and thus is incoherent outside of some utility reference frame.

7.2. Synthesis

Theorem (Completeness Requirement). A complete theory of power requires (although may be insufficient):

1. Relational framework ($\mathcal{R} \supset \mathcal{A}$)
2. Behavioral specifications for all agents (possibly without explicit utilities)
3. Action space definitions $\mathcal{A}_i$
4. Relevance constraints on state space $\mathcal{S}_i \subset \mathcal{S}$

Implication. Power analysis without game-theoretic foundations is fundamentally incomplete. $\square$

 Power of signaling. US vs Japan and communicating the existence of the nuclear bomb. US in some way doesn't have power over Japan if it doesn't benefit from either (A) Annihilating it or (B) Communicating its Annihilation.

References